University of Trento
Master's Degree in Computer Science

Data Visualization Proposal

Data Visualization Lab
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1 Context

1.1 Curiosity Origin

As frequent travelers, we are curious about urban mobility and how people navigate large cities. We became particularly interested in understanding mobility patterns in the context of sustainable transportation and green alternatives, recognizing their growing importance in urban planning and environmental goals. Specifically, we wanted to explore the underlying choices people make when selecting transportation modes and the factors that shape those decisions.

This curiosity led us to explore data sources that could provide meaningful insights into urban movement. Among them, New York City's bike rental data stood out as a rich source of information on how people travel across the city using a sustainable mode of transportation. When combined with other datasets, it can offer a clearer picture of the city's daily dynamics.

We also learned that this bike rental data is provided by a private company (`Citi Bike`) that operates the bike-sharing service in New York City. For this reason, making our analysis useful and relevant to the company is an effective way to increase its practical impact.

1.2 Project Circumstance and Purpose

Being one of the largest bike-sharing programs in the world, `Citi Bike` currently collects vast amounts of operational data. If effectively analyzed and visualized, this data could provide valuable insights to optimize their services and enhance user experience.

Our visualization tool directly addresses these gaps by making complex mobility data accessible and actionable.

The primary objective is to transform raw `Citi Bike` operational data into clear, actionable insights for stakeholders.

1.2.1 Citi Bike Operations Team

Specifically, we aim to reveal:

1. **Temporal patterns**: peak usage hours, seasonal trends, and demand cyclicity.
2. **Spatial hotspots**: high-demand stations, commute corridors, and underutilized infrastructure.
3. **User segmentation**: behavioral differences between members and casual riders.

4. **Rideable type preferences**: trends in rideable type usage (classic vs. electric) across different contexts.

The first two insights can help the operations team optimize station placement and bike rebalancing. Meanwhile, the last two can inform targeted marketing strategies and service improvements to better meet user needs.

This information reflects the most relevant angles we aim to explore, as it directly informs operational decisions and strategic planning for the company.

1.2.2 City Planners and General Public

Beyond the operations team, our visualization can also benefit city planners by providing insights into commute flows and modal patterns, which can inform infrastructure investment and urban planning decisions.

For the general public, it can raise awareness of urban mobility and sustainable transportation alternatives, fostering a more informed and engaged community.

2 Proposed Datasets

Before selecting the datasets used in this project, we explored several mobility datasets from different cities in order to identify a suitable data source. However, we found that many of these datasets were either incomplete, not regularly updated, or lacked the necessary granularity for our analysis.

After careful consideration, we decided to focus on the `Citi Bike` System Data for New York City, which is well maintained and reflects the complexity of the city.

2.1 Citi Bike System Data

The Citi Bike System Data is published by *Lyft*, a company that operates the bike-sharing service in New York City. Trip records, dating back to 2013, are released monthly as open data, and they are available for download in CSV format.

Each trip record includes key attributes such as start and end timestamps, start and end stations, and user type. These variables allow us to analyze both temporal and spatial patterns in bike usage.

We can also access real-time bike availability and station status via public APIs and integrate it into our dashboard for up-to-date insights.

2.2 Weather Data

Weather conditions significantly influence urban mobility. To better understand how factors such as temperature, precipitation, and wind affect ridership, we will incorporate historical weather data into our analysis.

We will utilize the Open-Meteo API, which provides historical weather data for any location worldwide, including New York City.

The dataset covers more than 80 years of observations with hourly temporal resolution and a spatial resolution between 1 and 11 km. As an open-source service, Open-Meteo offers a comprehensive set of variables, enabling us to better evaluate the impact of environmental conditions on the system.

2.3 New York City Datasets

During our exploration, we also considered various datasets available on the NYC Open Data platform. It offers a wide range of datasets, dashboards, and maps related to the city's transportation system, including bike lanes, traffic data, and other contextual information.

However, we want to keep our analysis focused on the bike-sharing system, and the `Citi Bike` System Data already provides a comprehensive view of the city's bike rental patterns. Therefore, we decided to avoid adding unnecessary complexity by incorporating additional datasets from NYC Open Data, unless they are needed to provide further context or support specific insights.

2.4 Datasets Limitations

Another important aspect concerns the quality and consistency of the available data. Historical `Citi Bike` records were collected over multiple years, and they may contain missing values, inconsistencies, or differences in formatting and schema across datasets. For this reason, a careful preprocessing phase is required to clean, standardize, and integrate the data before any reliable analysis can be performed.

Beyond simple cleaning, extracting new features is essential for uncovering meaningful insights. Transforming raw timestamps into categorical variables, calculating distance from coordinates or computing advanced statistics allows us to capture patterns that aren't immediately visible in the original dataset.

Then, the integration of weather data also introduces challenges, as it requires aligning granularity in both time and space with the bike trip records.

Finally, the large size of the dataset introduces additional technical challenges. Efficient aggregation, filtering, and data handling strategies must be

implemented to manage the volume of information effectively.

3 Solution

3.1 Vision

We deliver an interactive web-based dashboard as the primary output. This allows all users to explore mobility patterns through a unified, user-friendly interface. Our design approach is to present clear, immediate insights at first impactful while also allowing deeper exploration for users who want to examine the details.

Regarding the update frequency, we expect the dashboard to be refreshed automatically on a monthly basis, coinciding with the release of new data from the provider. This ensures that the insights remain useful and relevant over time, while also enabling historical analysis and comparisons across different periods.

Regarding framing, we aim to expose all available features, giving users the freedom to explore in depth and draw their own conclusions. However, we also provide some guidance through the design of the interface and the choice of default views.

3.2 Design Philosophy

Our visualization prioritizes clarity and analytical depth over visual polish. We employ a dual-level approach: high-level dashboards present immediate summary insights (e.g., daily ridership totals, top stations), while detailed views enable granular exploration (e.g., hourly breakdowns, user type comparisons). All design choices remain data-centric, avoiding decorative elements or misleading visual encodings that could obscure underlying patterns.

3.3 Early Design Considerations

While specific chart types and encodings will be refined during development, a few design decisions follow directly from the editorial choices above:

- **Map-centric spatial view**: Spatial data is inherently geographic, so an interactive map is the natural primary choice for our project. A heatmap overlay can show station-level demand, while flow lines can illustrate the most common routes between stations.
- **Faceted temporal charts**: Temporal data will be used mainly to compare behavior across different time periods, so faceted line charts or small multiples will allow us to compare patterns across days of the

week, hours of the day, or seasons without overwhelming the viewer with too much information in a single chart.

- **User and rideable type breakdowns**: Bar charts or stacked area charts can effectively show the distribution of user types and rideable types, supporting comparisons across different contexts (e.g., time of day, weather conditions).
- **Interactive filtering over static views**: Rather than producing separate dashboards for each dimension, we favour cross-filtering: selecting a time range, station, or user type in one chart updates the others, supporting exploratory analysis without overwhelming the interface.

Regarding focus, the map-centric spatial view serves as the primary entry point, offering the most intuitive and immediate reading of the data. Temporal and user/rideable breakdowns take a supporting role, enriching the analysis without cluttering the interface.

3.4 Constraints

In general, obtaining the most value from the data while respecting project constraints is a key challenge. The project must balance the ambition of performing meaningful analyses with practical limitations such as time availability, data quality, and implementation complexity. Since the final deliverable is scheduled for June, it is necessary to keep the feature scope focused and prioritize analyses that can provide the most relevant and actionable insights.

In order to manage both data processing and dashboard development within the project timeline, we want to leverage our programming skills to build a tech stack defined as follows:

- **Backend (Python + Polars)**: efficient data processing, cleaning, and REST API development via FastAPI;
- **Frontend (React + D3.js/Deck.gl)**: custom, interactive visualizations tailored to mobility pattern analysis. Deck.gl will be used for the map-centric spatial view, while D3.js will handle temporal and categorical charts.

This modular architecture enables rapid iteration during development while remaining scalable as data volumes grow.